Data-Driven Retrodiction with Red-String Maps and Causal Memory

Alternative pasts, uncertainty-aware smoothing and delayed-response reconstruction

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Abstract

Retrodiction infers earlier states or events from observations collected later. This paper develops a mathematical instruction framework that preserves complete alternative histories in a red-string map, uses temporal WPM landscapes as structured descriptors, and improves inference through uncertainty-aware smoothing and independently measured delayed responses. It derives branch ambiguity, backward-message recursions, signed-memory discrimination, temporal-curvature identities, observational aliases and whole-path uncertainty rules. The model learns transition and response relationships from permitted observations or a declared paired archive; it is not supplied the true generating equation. The treatment contains constructed mathematical examples and proofs, with no experimental result tables or empirical superiority claims.

Research lineage and mathematical status. This study develops a WPM mathematical extension with reference to the author's published WPM architecture [1] and foundational n/s scaling-spectrum framework [2]. The n/s paper concerns conditional moment-spectrum parameterization; the WPM paper provides the broader architectural reference. The results proved here are mathematical consequences of stated assumptions, usually applications of standard linear algebra, probability or optimization. The WPM contribution is their organization into an explicit workflow. Modeling constructions and applicability conditions are identified separately; no universal forecasting superiority is asserted.

Method provenance. The red-string branch representation and delayed-response causal-memory extension are formalized from the author's earlier retrodiction manuscript, including its causal-memory addendum [7]. The present study develops the mathematical rules and assumptions without reproducing that manuscript's experimental results.

Data-driven inference contract. The default retrodictor estimates transition, response and noise relationships from observed later data or a separately declared archive of paired earlier/later records. It is not fed the true generating equation, hidden earlier states, true response delays or true noise parameters. Model grammars, state domains, priors and assumed causal direction remain explicit restrictions. Supplied-equation examples below explain mathematical limits; they do not imply that the inference procedure knows the physical generator.

1 Define the missing past and the available later evidence

Let the unobserved target history be $X_- = X_{-H:-1}$ and the observed later record be $D_+ = \{Y_s, Z_s : 0 \le s \le L\}$. Here H is backward reconstruction depth and L is observed later extent. In WPM, W is a past lookback at a landscape snapshot, m is temporal sampling or resolution, and P is a forward probe horizon used to construct that landscape. These are different quantities; a long observed later record does not automatically supply a long historical anchor.

A retrodiction target may be the full path, a past event, a sign, an integral, or a time of occurrence. Write it explicitly as $\Phi(X_-)$. Identifying a selected functional can be possible when the complete state is not observable, a distinction related to functional observability [5]. A path estimate, a set of feasible histories and a probability of a past event are different outputs.

The inference information set is the actual later record available when the retrodiction is issued. Within that record, a landscape snapshot constructed at time s may use only probe outcomes matured by that snapshot, under its declared causal convention. The final retrodictor can use the full observed later interval; this is smoothing with later evidence, not a live forecast pretending to know future measurements.

An optional historical anchor is a separately measured record $AX_- = b + \epsilon_A$. State its time span, noise model and provenance. It adds observations and must not be hidden inside a supposedly later-data-only method. A candidate past drawn from an archive is not an anchor measurement of the query system.

2 The red-string map preserves complete alternative histories

Represent inference as a directed chronological graph. A state node stores physical time, values or distributions, branch identity, model version and observed/inferred status. A pocket node additionally stores its individual identity, support, flow curves, position, existence, signed channel behavior and provenance. Edges identify transition, archive match, measured anchor, inferred link, or approximate boundary join.

Several paths can share observed later nodes while retaining distinct earlier nodes. A stable branch ID lets the reader return to any earlier time or pocket without collapsing alternatives. An archive join is labeled approximate unless an exact transition constraint is verified. Merging nodes for storage does not authorize merging distinct evidential hypotheses.

Definition 2.1 (A retained red-string branch)**.** A branch b consists of a complete candidate path $x^{(b)}_{-H:L}$, its state and pocket records, its scoring assumptions, a support or weight record, and typed links to the actual observations. Hypothetical historical pockets inherit the status *inferred from candidate history*; they never become observed past evidence merely because they were plotted.

Branch navigation is a reproducible indexing operation. It does not establish that the indexed history occurred. Preserve rejected branches or their rejection reasons when practical. If pruning or finite sampling discards paths, record the rule and candidate budget; absence from the saved graph need not mean physical impossibility.

Worked example 2.2 (An average can be no valid history)**.** For the supplied illustrative map $x_{s+1} = x_s^2$, observation $x_0 = 1/4$ admits $x_{-1} = \pm 1/2$. The equal-weight mean predecessor is zero, but its forward image is zero, not $1/4$. A squared-error point summary can therefore fail the transition relation. Store both complete branches even when also reporting their mean.

3 What later data can identify

Theorem 3.1 (Indistinguishable histories)**.** *Let b_1, b_2 be two candidate histories with identical distributions for every measured later datum: $p(D_+ \mid b_1) = p(D_+ \mid b_2)$. Under a specified Bayesian model their posterior odds equal their prior odds whenever these are defined. No deterministic transform of the same later data can distinguish them statistically.*

Proof. Bayes' rule expresses posterior odds as prior odds times the likelihood ratio, which is one. For a deterministic transform $T(D_+)$, its distribution is the pushforward of the same data law under both histories and is therefore identical. □

In a deterministic Markov system, histories that merge into the same complete state at the observed boundary have the same future if the same later forcing is applied. Additional future observations of that state alone cannot distinguish the predecessors. But a visible scalar may omit a memory coordinate: distinct pasts can merge in visible x_0 while leaving different delayed responses. The complete state then has not merged.

Proposition 3.2 (Extra information cannot increase optimal expected squared error)**.** *For square-integrable target $U = \Phi(X_-)$, information sets $\mathcal{G} \subseteq \mathcal{H}$, and conditional-mean estimators,*

$$\mathbb{E}\,\|U - \mathbb{E}[U \mid \mathcal{G}]\|^2 = \mathbb{E}\,\|U - \mathbb{E}[U \mid \mathcal{H}]\|^2 + \mathbb{E}\,\|\mathbb{E}[U \mid \mathcal{H}] - \mathbb{E}[U \mid \mathcal{G}]\|^2 . \tag{3.1}$$

Proof. Decompose the first error into $U - \mathbb{E}[U \mid \mathcal{H}]$ and the difference of the conditional means. The cross term has expectation zero after conditioning on $\mathcal{H}$. □

This is an oracle information result averaged under a correct joint law. A finite fitted smoother can worsen under misspecification or population shift. It also does not imply improved error for every individual case or perfect branch identification.

4 Temporal pockets organize the observed evidence

At several later snapshot times s_j, compute matured probe-loss surfaces over (W, m, P) and retain their full scalar values where available. Identify connected useful basins under an explicit adjacency and threshold. Preserve each pocket's existence, position, full flow cross-sections and signed responses before its trajectory. Retain separate regimes and their relational graph: spacing, prominence relationships, scale ratios, births and disappearances can matter even when no one pocket moves far.

Use consistent units and record local normalizers. Fit comparison scales only from permitted archive or development data. Pocket IDs have no ordinal meaning; match their interpretable coordinates or use a permutation-invariant set representation. Signed channel responses must remain distinguishable from unsigned magnitudes, especially when the missing variable is a sign.

If $T(D_+)$ is the deterministic pocket representation, then

$$p(X_- \mid D_+, T(D_+)) = p(X_- \mid D_+) \tag{4.1}$$

under the same complete probabilistic model. Pockets can improve numerical conditioning, learning efficiency, retrieval and explanation. They are not an additional independent sensor. Multiplying a raw-data likelihood by a second likelihood for the same deterministic transform double-counts evidence unless a valid joint observation model is supplied.

Candidate historical landscapes can be computed from candidate histories, but any window requiring unavailable states must be marked unsupported or integrated over explicit latent states.

Do not pad a short reconstruction to manufacture a long historical pocket. Labels inferred from a branch may help describe that branch; they cannot be recycled as new measured constraints favoring it.

5 Paired-archive retrieval as a learned prior route

Suppose a declared archive contains paired pasts $x_-^{(i)}$ and later records $D_+^{(i)}$. Define standardized distances between query and archive later histories, possibly using raw observations, temporal pockets and known anchors:

$$d_i = d_{\text{raw}}(D_+, D_+^{(i)}) + \lambda d_{\text{pocket}}(T(D_+), T(D_+^{(i)})) + \lambda_A d_A(b, Ax_-^{(i)}). \tag{5.1}$$

The anchor term exists only if query anchors are supplied. Select metric scales, λ, candidate count and any kernel temperature using separate known-history development records. Preserve raw-only and pocket-only controls; do not assume that their combination adds information beyond the full raw record.

Weights $w_i \propto \exp(-d_i/\tau)$ are normalized compatibility weights unless the kernel, candidate prior and observation process define a justified likelihood. Top-K retrieval omits other histories. It cannot identify the probability mass of unseen candidates or prove that the query's true past lies in the archive.

For full-path squared loss, $\sum_i w_i x_-^{(i)}$ is a point summary under the weighted empirical model. An actual selected branch can instead be its weighted medoid, minimizing $\sum_i w_i d(x_-^{(i)}, x_-^{(b)})$ over saved branches. These minimize different constrained objectives and should not be presented as interchangeable.

6 Learn dynamics from later observations, then invert cautiously

An alternative route fits a transition model $x_{s+1} \approx f_\theta(x_s)$ from observed later pairs, with a measurement model for noisy Y_s. A polynomial, neural map or other family is a declared grammar. Fit and select it using permitted observations; fitting a polynomial directly to noisy regressors can bias its coefficients. A latent-state treatment or a measurement-error correction may be required.

For a noiseless supplied illustrative fold $f(x) = 1 - ax^2$, $a > 0$, predecessors of $v \le 1$ are

$$x = \pm\sqrt{(1-v)/a} \tag{6.1}$$

when real and within the declared state domain. Repeated inversion retains every feasible predecessor path. Two roots do not mean two equally probable pasts unless a probability model makes them so. A finite domain, unknown earlier forcing or a different earlier transition law can change admissibility.

Under continuous densities, a noiseless many-to-one transformation also carries a Jacobian. For $v = f(x)$ with simple roots x_i and prior density p_X,

$$\mathbb{P}(X = x_i \mid f(X) = v) \text{ as a branch weight } \propto \frac{p_X(x_i)}{|f'(x_i)|}. \tag{6.2}$$

This regular-conditional branch formula follows by change of variables locally; it is not an ordinary positive probability of an exact continuous point. At critical roots it requires separate limiting treatment. A likelihood over noisy observations avoids treating a measured noisy anchor as an exact inverse boundary.

7 Uncertainty-aware smoothing on complete paths

Declare a transition kernel $K_\theta(x' \mid x)$, prior π_θ and observation likelihoods. Transition discrepancy represents uncertainty or model error; it must not be interpreted as known physical process noise without evidence. A working trajectory model has factors

$$p(x_{-H:L}, \theta \mid D_+) \propto p(\theta)\pi_\theta(x_{-H}) \prod_{s=-H}^{L-1} K_\theta(x_{s+1} \mid x_s)\, p(D_+ \mid x_{-H:L}, \theta). \tag{7.1}$$

The displayed distribution is a posterior only under the declared generative assumptions; estimated surrogates may instead define a working smoothing distribution. Integrate parameter uncertainty when feasible. A finite coefficient ensemble approximates uncertainty and is not automatically an exact posterior.

For a finite-state or quadrature approximation with local likelihood $\ell_s(x_s)$, define backward messages

$$\beta_L(i) = 1, \qquad \beta_s(i) = \sum_j K_s(j \mid i)\ell_{s+1}(j)\beta_{s+1}(j). \tag{7.2}$$

The smoothed boundary weights are proportional to $\pi(i)\ell_{-H}(i)\beta_{-H}(i)$. Generate complete paths using conditional transitions proportional to $K_s(j \mid i)\ell_{s+1}(j)\beta_{s+1}(j)$. Retain normalizing constants, quadrature weights and log-scaling for numerical stability. Grid resolution, domain truncation and path budgets require sensitivity checks.

Proposition 7.1 (Backward-message identity)**.** *Under the local Markov factorization, $\beta_s(i)$ equals the likelihood of observations strictly after s conditional on $x_s = i$. The resulting path sampler draws from the finite-model smoothing distribution.*

Proof. At L there are no later factors. Condition on $x_{s+1} = j$, multiply its transition, local observation likelihood and remaining future likelihood, then sum over j. Induction proves the recursion. Dividing each factorization by its conditional normalizer gives the stated path transitions. □

For continuous systems, particle methods or other smoothers provide alternatives; particle Gibbs with ancestor sampling is one established trajectory-sampling method [6]. A finite-grid message calculation should not be mislabeled as that algorithm.

8 Correlated noise, robust likelihoods and soft anchors

Temporally correlated residuals do not generally factor into independent one-time Gaussian densities. For an AR working model $e_s = \rho e_{s-1} + \nu_s$, retain the innovation likelihood and initial-residual distribution, or augment the state. A heavy-tailed innovation model can reduce outlier influence but does not solve arbitrary model drift. If the two channels share noise, use their joint covariance or another justified joint likelihood rather than multiplying independent-channel factors.

A noisy anchor contributes $p_A(b \mid Ax_-)$, such as a Gaussian with declared covariance. Exact filtering $Ax_- = b$ is appropriate only for genuinely exact constraints. Approximate boundary matches, uncertain measurements and uncertain fitted transitions should remain soft. An empty candidate set diagnoses conflict or computational failure; it is not evidence that no physical past existed.

Parameter selection and uncertainty qualification need known-history development and calibration cases distinct from the unknown query. If they are unavailable, label resulting support as model-conditional. A narrow smoother under the wrong earlier law can confidently miss the true history.

9 Causal memory: later effects of missing earlier states

The additional response channel can have the illustrative form

$$Y_s = x_s + \epsilon_s^Y, \qquad Z_s = c + b\, g(x_{s-d}) + \epsilon_s^Z, \quad d > 0, \tag{9.1}$$

where $g(x) = x$ for signed memory or $g(x) = x^2$ for energy-like memory. Inference estimates b, c, d, g from later observations for which both cause and response are observed, under the candidate grammar. Assumed causal direction is a restriction; correlation alone does not rule out confounders or shared measurement noise.

For $0 \le s < d$, response Z_s concerns an earlier state x_{s-d}. These measured responses can discriminate candidate pasts even when state-only later data do not. There is no reversal of physical causation: an earlier cause leaves a later measured trace. The effective state includes the delay buffer $S_s = (x_s, x_{s-1}, \dots, x_{s-d})$, or a valid reduced sufficient representation.

Direct memory reach among missing states is the set of $s - d < 0$ for responses actually observed. Observing through L covers direct causes from $-d$ through $\min\{-1, L-d\}$. A history deeper than that requires dynamical constraints or additional data; nominal delay alone does not establish complete reach when response measurements are missing.

Worked example 9.1 (Signed memory resolves a two-branch fold)**.** Suppose $x_{-1} = \pm r$, $r = .5$, are equally plausible from state-only later data. An observed signed response is $z = bx_{-1} + \epsilon$, with stipulated known $b = .6$ and $\epsilon \sim \mathcal{N}(0, \sigma^2)$, $\sigma = .2$. Then

$$\log \frac{p(z \mid +r)}{p(z \mid -r)} = \frac{2brz}{\sigma^2}. \tag{9.2}$$

For $z = .3$, the log odds are 4.5 and the positive-branch probability under the equal-prior model is $\sigma(4.5) \approx .9890$. Here $\sigma(\cdot)$ denotes the logistic function, distinguished from the noise standard deviation by its argument. In a data-driven application uncertainty in estimated b must also be propagated.

Proof. Subtract the Gaussian squared residuals: $[(z + br)^2 - (z - br)^2]/(2\sigma^2) = 2brz/\sigma^2$. Bayes' rule supplies equal-prior odds. □

An energy response $z = c + bx^2 + \epsilon$ gives identical likelihoods for $\pm r$. An unknown response sign with a symmetric prior can also preserve the joint symmetry $(x, b) \mapsto (-x, -b)$. More precision in a sign-blind measurement does not resolve a sign ambiguity. Signed, separately measured information or another symmetry-breaking restriction is required.

10 Joint generation is stronger than reweighting a fixed archive

Given raw candidates $\{x_-^{(b)}\}$ and valid response likelihood, reranking multiplies their weights by $p(Z_+ \mid x_-^{(b)}, Y_+)$. If no candidate is close to the true past, reranking cannot create it. A joint smoother uses the response channel during candidate generation as well as selection, and can explore regions omitted by a raw-only proposal.

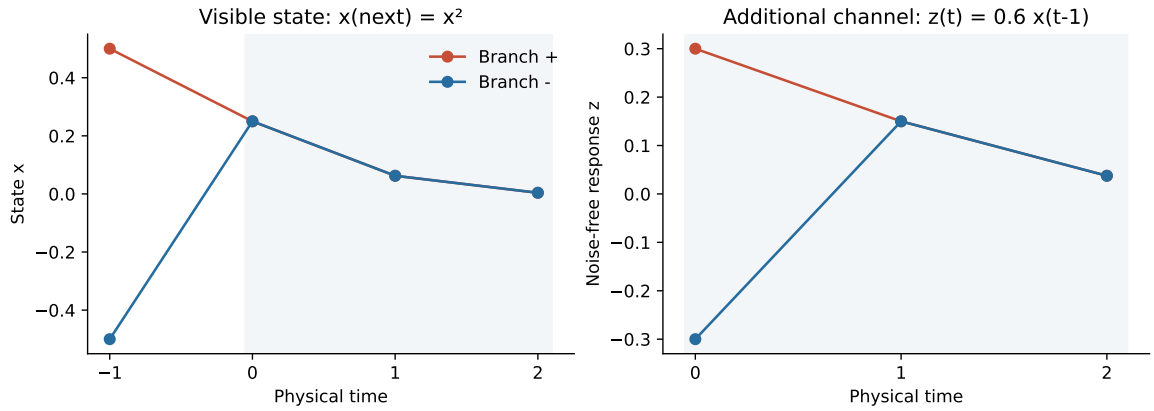


Figure 1: Constructed square-map branches merge in the observed scalar state, while a one-step signed-memory channel retains different boundary responses. Measuring that additional channel changes the observation problem. Neither colored branch is labeled historical truth.

For independent channel-noise processes conditional on the complete latent path, the state and response factors multiply. Otherwise use the joint likelihood. A delayed factor can be attached to its earlier cause when coefficients and delay are fixed, but temporal response-noise correlation may require memory augmentation or a nonlocal factor graph. Do not apply the simple scalar recursion while silently dropping those dependencies.

Model components may include transition parameters, response operator, delay, noise law and earlier response polarity. Integrate or retain component uncertainty. A selected best operator is a point estimate, not proof of its unique physical identity. Normalize component evidence consistently, including every prior and boundary factor.

11 Temporal curvature and an earlier response law

The missing segment may have response coefficient b_- different from the later fitted coefficient. To explain how a temporal trace can constrain it, consider the ideal noiseless model

$$x_{j+1} = \theta_0 + \theta_1 x_j + \theta_2 x_j^2, \qquad r_j = c + b x_j, \quad b \neq 0. \tag{11.1}$$

This is a derivation under a shared state law and a segment-constant response map, not an assumption that inference receives the true coefficients.

Theorem 11.1 (Affine response conjugacy and curvature)**.** *The response satisfies* $r_{j+1} = \alpha_0 + \alpha_1 r_j + \alpha_2 r_j^2$ *with*

$$\alpha_2 = \theta_2/b, \tag{11.2}$$
$$\alpha_1 = \theta_1 - 2\theta_2 c/b, \tag{11.3}$$
$$\alpha_0 = c + b\theta_0 - \theta_1 c + \theta_2 c^2/b. \tag{11.4}$$

If θ_2 *and* α_2 *are known and nonzero, then* $b = \theta_2/\alpha_2$ *and* $c = (\theta_1 - \alpha_1)/(2\alpha_2)$*. These algebraic identifications require enough informative observations to identify the response polynomial.*

Proof. Substitute $x_j = (r_j - c)/b$ into $c + b f_\theta(x_j)$ and collect constant, linear and quadratic powers of r_j. Solve the nonzero quadratic coefficient for b and the linear coefficient for c. □

For the fold $x_{j+1} = 1 - a x_j^2$, curvature is $-a/b$. Reversing b reverses curvature. A single isolated response may leave sign ambiguity, while a temporal trace subject to a shared nonlinear law may break it.

Worked example 11.2 (Curvature worksheet)**.** With $\theta = (1, 0, -2)$, $b = .5$ and $c = .1$, the response coefficients are $(\alpha_0, \alpha_1, \alpha_2) = (.56, .8, -4)$. Inverting gives $b = .5, c = .1$. Direct substitution at $x = .3$ gives $r = .25$, $x' = .82$ and $r' = .51$; the response polynomial yields $.56 + .8(.25) - 4(.25)^2 = .51$.

Noisy response-polynomial fitting is an errors-in-variables problem. Latent smoothing or a justified measurement model is needed before treating estimated curvature as evidence of an earlier coefficient. If the state-law curvature $\theta_{2,\text{past}}$ is also unconstrained, the trace identifies the ratio $\theta_{2,\text{past}}/b_-$, not those quantities separately. Weak response amplitude, short traces and narrow state range make identification unstable.

12 Polarity mixtures, source aliases and historical changes

A constrained repair can retain two earlier-polarity hypotheses $b_- = \kappa b_+$, $\kappa \in \{-1, +1\}$, with explicit prior weights and a shared canonical offset. This allows a particular boundary change without asserting recovery under arbitrary earlier equations. If the response changes again within the missing window, a one-boundary mixture is misspecified. More change points require their own priors, evidence and complexity control.

For a fold $x_j = 1 - ax_{j-1}^2$, a signed response at one time has an energy alias:

$$c + bx_j = (c + b) - abx_{j-1}^2. \tag{12.1}$$

Thus signed delay d can describe the same response as energy delay $d + 1$ with transformed intercept and slope. A response fit may identify an observational equivalence class rather than one unique operator/delay label. Negating an energy coefficient without transforming its intercept generally does not implement the intended signed-polarity reversal.

Report separately the recovered trajectory functional, the response prediction, and the structural equation label. Accurate response reconstruction does not establish unique causal mechanism. Independent measured channels, known invariants or historically justified restrictions may break an alias; selecting a preferred label does not.

13 Past-event probabilities, branch selection and calibrated sets

For a past event $E = \{\Phi(X_-) > b\}$, a valid working path distribution yields $p_E = \sum_i w_i \mathbf{1}\{\Phi(x_-^{(i)}) > b\}$. Label its model and approximation. Compatibility weights alone give a weighted support fraction, not a calibrated physical probability. An event can be identifiable even with multiple paths if every retained feasible path agrees; it can also be unidentifiable despite a very precise endpoint.

For selected path $\widehat{x}_-$, local mass $\sum_i w_i \mathbf{1}\{d(x_-^{(i)}, \widehat{x}_-) \le \delta\}$ describes working-model concentration. It is different from the probability of exact sign sequence, a frequentist accuracy guarantee, and the chance that at least one candidate is close to truth. Finite candidate budgets can make apparent concentration misleading.

To qualify complete-path candidate sets on exchangeable independent known-history cases, define

$$S_i = \min_{b \in \mathcal{B}(D_i)} \max_{-H \le s < 0} |x_{i,s} - x_{i,s}^{(b)}|. \tag{13.1}$$

Use $S_i = +\infty$ for an empty set. With n independent calibration cases, let $k = \lceil (n+1)(1-\alpha) \rceil$ and choose the kth order statistic q, including the usual $+\infty$ convention when $k = n + 1$. For

finite q, the output is a union of whole-path tubes, one tube around each complete branch; an empty candidate set then gives an empty prediction set. If $q = +\infty$, explicitly return the full admissible path domain, including for an empty candidate set. With that fallback, standard exchangeable rank reasoning yields marginal coverage at least $1 - \alpha$, conditional on a previously fixed candidate-generation procedure [4]. Randomized candidate generation must be included consistently in that procedure. A pipeline that always leaves empty sets empty cannot claim this unrestricted guarantee.

For $n = 19$ and nominal coverage .9, the rank is 18. Coverage applies to the calibrated population, not automatically every noise regime, subfamily, or historical-law shift. Report set size, tube width and empty-set frequency alongside coverage. For a fixed δ, the union of path tubes is not the Cartesian product of their coordinate-wise unions: choosing a different branch at each time can create an invalid history.

14 An explicit retrodiction procedure

1. Declare missing times, backward depth, target functional, observed later extent, channels and optional measured anchors. Separate raw observations from audit-only historical truth.
2. Preserve complete later histories and temporal pocket records, including signed responses, flow, position and existence. Keep independent regimes and pocket relationships.
3. Choose a paired-archive or learned-state-model route. State grammars, domains, priors, boundary assumptions and parameter fitting data.
4. Search for independently measured delayed responses using later times where causes are observed. Account for the search over operators and delays, measurement noise and shared-noise alternatives.
5. Generate complete branches under uncertainty. Use soft anchors, model uncertainty and response memory where supported. Retain null-response and state-only branches as comparators.
6. Preserve aliases and alternative earlier laws when identifiable restrictions permit them. Distinguish reach from correctness of the transported response law.
7. Select an actual branch if required, while retaining alternatives and path-level uncertainty. Compute past-event support under an explicitly labeled weighting model.
8. Qualify selected-path error and whole-path sets on independent known histories. Report abstention coverage and failure under unsupported earlier laws. Save the red-string graph for exact retrieval.

These instructions improve the formulation of the inverse problem; they do not guarantee information that was never measured. A delayed measured effect can add information. A new name, a pocket transform or a larger branch map alone cannot.

15 Proof trail and solved reconstruction questions

The indistinguishability theorem uses a likelihood ratio and pushforward distributions. Expected-risk improvement uses orthogonal conditional expectations. Backward smoothing follows a Markov factorization. Signed-memory discrimination is a Gaussian likelihood ratio. Curvature recovery is polynomial conjugacy. Alias handling is substitution. Whole-path qualification uses exchangeable ranks. Each claim depends on its stated observation and model assumptions.

Exercise 15.1. Two candidate histories reach the same complete boundary state and induce identical later measurement distributions. Can more pocket snapshots computed from those measurements reveal which history occurred?

Solution. No. A deterministic representation preserves the likelihood equivalence. An additional measured memory channel could help only if its distribution differs across those histories.

Exercise 15.2. The state-only candidates omit a past compatible with a strong delayed response. Can reweighting the existing candidates recover that missing branch?

Solution. No. Reranking changes weights, not support. Joint candidate generation or a broader proposal is needed, followed by independent qualification.

Exercise 15.3. An energy response observes only x^2 at a fixed magnitude. Does reducing measurement noise identify the sign?

Solution. No. Both signs have the same response likelihood. A symmetry-breaking measurement or justified dynamic restriction is needed.

Exercise 15.4. May a selected path's local working mass of .95 be reported as independently verified 95% reconstruction accuracy?

Solution. No. It is concentration under the working model. Accuracy requires an independently evaluated correctness event, stated tolerance and supported target population.

Scope of the mathematical exposition

Unless explicitly identified as a sourced evidence statement, numerical values are constructed arithmetic examples. Application requires specifying the observation process, fitting unknown quantities, and checking reliability for the intended population and horizon. The archival DOI for this study is 10.5281/zenodo.23785161.

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